Dr. Sathyanarayan Rao

Research Scientist & Computational Engineer

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Education

- **PhD in Engineering Sciences**, UCLouvain, Belgium 2016 – 2020
  Thesis: Computational Modeling of Electrical Signatures of Plant Roots
  Awarded prestigious FNRS fellowship for doctoral research.
- **MS in Applied Physics**, Alabama A&M University, USA 2013 – 2014
  GPA: 4.0/4.0
  Conducted laboratory research in laser-matter interaction and photodegradation.
- **MS in Electrical Engineering**, University of Alabama in Huntsville, USA 2010 – 2012
  GPA: 3.9/4.0
  Research funded by NSF grant ATM0647157 on plasma simulation and modeling.
- **B.Eng in Electronics and Communication**, VTU, India 2006 – 2010
  First Class with Distinction (77%, equivalent GPA: 3.9/4.0)
  External project at Bharat Electronics Limited (BEL) in digital communication systems.

Experience

- **Scientific Coding & Applied Mathematics Consultant**, Turing (Client: Frontier AI Research Organization) January 2026 – Present
  - Engaging as a consultant through Turing to support a leading AI research organization on **next-generation large language model training and evaluation**.
  - Designed and analyzed **scientific coding and applied mathematics tasks** to identify systematic failure modes in model reasoning, with emphasis on where **heuristic pattern-matching diverges from mathematical necessity**.
  - Investigated **reasoning bifurcation points** in areas such as linear algebra, numerical methods, graph theory, statistical inference, and dynamical systems, distinguishing algorithmic derivation from parametric recall.
  - Developed adversarial-yet-valid benchmarks exposing vulnerabilities related to **indexing conventions, coordinate reconstruction, semantic overload, equilibrium bias, and invariant suppression**.
  - Contributed to improving model robustness by ensuring correct handling of **edge cases, ambiguous scientific language, and structurally underdetermined problem statements**.
  - Work conducted under strict confidentiality agreements; no proprietary model details, datasets, or internal methodologies disclosed. - Additionally contributing to the **Multi-modal**, designing PhD-level image-dependent STEM benchmarks targeting failure rates to identify limits of AI visual and scientific reasoning.
- **Research Associate**, Indian Institute of Science (IISc), Bengaluru April 2025 – Present

- Designed and implemented **machine learning models (Random Forest, time-series analysis)** to improve SAR-based soil moisture estimation, achieving **34% accuracy gain** for sandy textures.
- Analyzed **multi-year time-series data** (soil temperature and moisture at multiple depths) to uncover patterns in groundwater recharge, soil-water balance, and climate-driven disruptions.
- Developed **automated pipelines in Python** for data cleaning, feature engineering, and statistical diagnostics (van Genuchten modeling, gradient analysis, recharge event detection).
- Building **interactive dashboards** for visualizing soil–moisture–temperature dynamics, supporting both researchers and policymakers.
- Designed a **staggered BACI Difference-in-Differences (DiD) causal framework** using a decade of Sentinel-1 SAR time-series (stratified sample points, 7-stage GIS spatial filtering, 5-stage radiometric processing) to isolate watershed intervention effects from rainfall variability across two paired semi-arid catchments in Karnataka, isolating a **+19.7% soil moisture treatment effect** ($p < 0.001$) that exceeded World Bank targets for all four soil texture classes.
- Validated radar-derived findings with TDR depth measurements (10–80 cm) confirming root-zone infiltration during drought, and demonstrated **6-year structural persistence** via event-study analysis — framework now proposed for scaling across 20 watersheds and 200,000 hectares as a climate-robust monitoring tool for public watershed investment.

- **Research Software Engineer**, Phenorob Project, Forschungszentrum Jülich 2023 – 2025
  - Developed robust interfaces to couple one-dimensional crop models (C/C++) with plant-scale Functional-Structural Plant Models (FSPMs; Fortran/CPlantBox), implementing loose-coupling data exchange, timestep synchronization and bidirectional parameter passing.
  - Created a python script for mechanistic sink-term for AgroC based on parallel root models enabling realistic root water uptake driven by dynamic root architecture (RLD) from CPlantBox.
  - Built a cross-platform AgroC GUI using PyQt5 to generate and validate ‘plants.in‘ input files, including tabular editors, dynamic table sizing and robust save/load routines to support model runs and scenario testing.
  - Packaged modelling tools and workflows with multiple Dockerfiles (AgroModeling stacks) enabling large-scale simulations on HPC clusters (batch scripts, container orchestration), improving reproducibility and scalability.
  - Implemented project website and documentation (Hugo, SCSS, JavaScript) and created scientific visualizations/animations to communicate complex soil–plant processes to researchers and stakeholders.
  - Led monthly project coordination, contributed to book chapters on LLMs and UAVs in agriculture, and authored a review on model coupling & digital-twin architectures (submitted to In Silico Plants).

- **Research Associate**, Indian Institute of Science, Bengaluru 2022 – 2023
  - Transitioned from computational modeling to hands-on field research, leading bi-weekly campaigns in the Berambadi watershed and standardizing data formats from collection to analysis.
  - Developed ML models (Random Forest, SVM, LSTM) for soil moisture prediction from SAR backscatter with ground-truth validation, co-presented poster at IIT Bombay Soil Moisture Workshop.
  - Built preliminary data pipelines in Python for cleaning and block-wise extraction of soil measurements; automated dashboards using Vue.js, Plotly, and Leaflet for interactive visualization.

- Trained students in instrumentation and protocols, ensuring reproducibility and quality of time-series datasets.
- Created short educational videos on soil moisture retrieval methods, supporting outreach and team training.
- Explored computer vision methods for leaf area index (LAI) estimation from smartphone imagery; gained experience with image analysis workflows, though accuracy was limited compared to professional (LiCor).

- **Independent Research & Development Consultant** 2020 – 2022
  - Returned to India during the COVID-19 pandemic while pursuing postdoctoral opportunities and academic grant proposals.
  - Developed Python-based analytical tools for financial market analysis and algorithmic trading strategies using WebSocket APIs.
  - Created machine learning pipelines for healthcare datasets and designed interactive UIs with frameworks such as Gradio.
  - Applied time-series modeling and signal processing techniques across finance, healthcare, and environmental datasets.
  - Demonstrated adaptability and resilience by maintaining active skill-building, and collaborations during the global disruption.

- **FNRS Research Fellow**, eRoot Project, UCLouvain, Belgium 2016 – 2020
  - Developed **finite element models (FEM)** with up to 500,000 tetrahedral elements to simulate soil–root electrical interactions at anatomical and field scales.
  - Integrated **root water uptake dynamics** and polarization effects into geoelectrical models, advancing non-invasive methods for root phenotyping.
  - Quantified **electrical anisotropy signatures** of root systems and linked them to structural indices of root architecture.
  - Applied **machine learning (k-nearest neighbors, PCA, statistical classifiers)** to discriminate root species from geophysical signatures.
  - Built Python-based automation pipelines to generate and process thousands of 3D root meshes (C-Rootbox + Gmsh) for simulation and validation studies.
  - Deployed simulations on **HPC clusters** (FZ Jülich supercomputing facility) for large-scale computational experiments.
  - Conducted field-scale Electrical Resistivity Tomography (ERT) experiments to phenotype root water uptake under drought stress, combining **numerical modeling with sensor data (ERT, TDR)**.
  - Published peer-reviewed journal articles and delivered conference presentations on computational agrogeophysics and non-invasive plant imaging.

- **Research Assistant** 2010 – 2016

  University of Paderborn, Germany 2015 – 2016
  - Developed **FORTRAN-based numerical solvers** to simulate excitonic resonances in two-level systems for optical communication applications.
  - Implemented **Maxwell–Bloch equations** using high-order Runge–Kutta, finite difference, and adaptive time-stepping for solution stability.
  - Modeled photon–matter interactions in engineered two-level systems under **DFG-funded optical signal processing research**.

  Alabama A&M University, USA 2013 – 2014
  - Conducted experimental and theoretical research on **laser-induced photodegradation** of dye molecules (Rhodamine 6G).

- Performed **photo-patterning and photo-deposition** of biosensing molecules on polymer thin films in cleanroom facilities.
- Operated and maintained high-value **laser systems, AFM, and precision optical benches**, ensuring nanometer-scale alignment.
- Analyzed biomolecule deposition patterns using **Atomic Force Microscopy (AFM)** with calibration and probe handling expertise.
- Served as **Teaching Assistant for Physics 101**, supporting laboratory sessions and student coursework.

University of Alabama in Huntsville, USA 2010 – 2013
- Extended 2D electrostatic **Particle-in-Cell (PIC) plasma code** into a fully functional 3D electromagnetic simulation framework.
- Developed **Helmholtz coil field generation modules** from scratch for plasma thruster digital twins under NSF funding.
- Captured key plasma dynamics (**wave–particle interactions, plasma instabilities**) in high-fidelity numerical models.
- Implemented parallelization with **MPI/OpenMP** for scaling on university HPC clusters.
- Contributed to peer-reviewed publications in *Physics of Plasmas* via low-level code development and Linux automation.

Teaching & Communication

- **Content Creator – Compute Stories (YouTube)** 2024 – Present
  - Produce educational and scientific communication videos simplifying complex concepts in **data science, computational modeling, and technical writing**.
  - Built a small channel `Compute Stories` channel, making advanced research topics accessible to students and professionals.
  - Skilled in **video editing (Final Cut Pro, DaVinci Resolve), 2D animation (Pencil2D)**, and **visual storytelling** to create engaging technical content.
  - Developed strong **presentation, communication, and outreach skills**, bridging the gap between academic research and industry applications.

- **Workshop Instructor – UCLouvain, Belgium** 2018
  - Designed and delivered a hands-on MATLAB workshop for **50 undergraduate students**, focusing on parameter optimization for soil hydraulic modeling.
  - Guided students in using non-linear least squares fitting techniques to estimate van Genuchten parameters from laboratory-measured datasets.
  - Managed two batches of students, providing one-on-one support, verifying results, and ensuring participants successfully completed computational hydrology exercises.
  - Strengthened **technical training, mentoring, and classroom management skills** through interactive problem-solving sessions.

- **Graduate Teaching Assistant – Alabama A&M University, USA** 2013 – 2014
  - Taught year-long **Physics 101 and 102**, covering mechanics, electrostatics, and other fundamentals of physics.
  - Conducted weekly laboratory sessions, training students in experimental methods and ensuring hands-on learning outcomes.
  - Designed course materials, graded assignments, and provided academic mentoring to students from **diverse cultural and academic backgrounds**.

- Gained practical experience in **inclusive teaching, student engagement, and cross-cultural communication**.

Invited Talks & Guest Lectures

- **A Classical Computational Person Looks at Quantum Computing: A Computational Scientist's View of Quantum Computing.** Invited guest lecture, Department of Electronics and Communication Engineering, **Kumaraguru College of Technology (KCT)**, Coimbatore, India, *October 2025*. Talk connected the mathematical and computational foundations of classical modeling, especially matrix-based systems with emerging quantum algorithms, emphasizing continuity between numerical simulation, linear algebra, and quantum computation.
- **Combining Sentinel-1 SAR Analytics and GPR Field Data for Subsurface and Infrastructure Assessment.** Invited technical presentation, Workshop on Advanced Non-Destructive Evaluation Techniques for Concrete Structures and GPR-Based Subsurface Investigations, **Kumaraguru Centre for Industrial Research & Innovation (KCIRI)**, Kumaraguru College of Technology, Coimbatore, India, *March 2026*. Presented a multi-scale assessment approach combining satellite-based Sentinel-1 SAR time-series analysis with ground-based GPR investigations for infrastructure monitoring, subsurface assessment, and identification of hydrological and structural interventions.

Publications

Journal Articles

1. Rao, S., Gupta, V., Murugesan, S., Zribi, M., & Muddu, S. (2026). An information proxy framework that explains why soil-specific radar calibration succeeds in sandy textures but fails in clay. *Remote Sensing Letters*, 17(4), 416–428.
2. Rao, S., Ahmadi, S. H., Leitner, D., Vianna, M., Bauer, F. M., Seidel, S. J., & Schnepf, A. (2026). Integrated modeling approaches for agricultural digital twins: the role of process-based models, agent-based models, machine learning, and model coupling. *In Silico Plants*, 8(1), diag002. https://doi.org/10.1093/insilicoplants/diag002
3. Rao, S., et al. (2026). High-frequency soil hydrothermal dynamics in a semi-arid agricultural catchment: a decade of 15-minute resolution observations (2016–2025). Submitted to *Earth System Science Data*.
4. Rao, S., et al. (2020). Imaging plant responses to water deficit using electrical resistivity tomography. *Plant and Soil*, 29 citations.
5. Rao, S., et al. (2020). Sensing the electrical properties of roots: A review. *Vadose Zone Journal*, 19(1), 71 citations.
6. Rao, S., et al. (2019). Impact of maize roots on soil–root electrical conductivity: A simulation study. *Vadose Zone Journal*, 18(1), 35 citations.
7. Singh, N., Rao, S., et al. (2013). Waves generated in the plasma plume of helicon magnetic nozzle. *Physics of Plasmas*, 20(3), 27 citations.
8. Rao, S., Singh, N. (2012). Numerical simulation of current-free double layers created in a helicon plasma device. *Physics of Plasmas*, 19(9), 39 citations.

9. Singh, N., Rao, S. (2012). Plasma turbulence driven by transversely large-scale standing shear Alfvén waves. *Physics of Plasmas*, 19(12), 3 citations.

10. Rao, S., et al. (in preparation). Detecting Watershed Intervention Effects on Soil Moisture: A Staggered Difference-in-Differences Approach Using Sentinel-1 SAR. Intended submission: *Remote Sensing of Environment*.

Books / Monographs

1. Rao, S. (2025). *Digital Twins: From Apollo Missions to Smart Farming*. Notion Press. ISBN: 979-8900231457. (Available on Amazon: https://www.amazon.in/dp/B0FP5JS1KL)

Book Chapters

1. Rao, S., Ranganath, P. (2025). Climate-Resilient Agriculture: Leveraging Language Models for Mitigation and Adaptation. In *Mitigation and Adaptation Strategies Against Climate Change in Natural Environments*.

2. Rao, S., Ranganath, P. (2025). Unmanned Aerial Vehicles (UAVs) as Sensor-Driven Data Feeders for Agricultural Digital Twins. Book chapter (submitted to Taylor & Francis).

Conference Presentations

1. Rao, S., et al. (2023). Random Forest for Soil Moisture Retrieval using Satellite Backscatter. Poster presented at *Soil Moisture Workshop*, IIT Bombay.

2. Rao, S., et al. (2019). Investigation of electrical anisotropy as a root phenotyping parameter: Numerical study with root water uptake. *Geophysical Research Abstracts*, 21.

3. Rao, S., et al. (2019). Relationship between electrical anisotropy of soil-root continuum and geometrical architecture of root system. *National Symposium for Applied Biological Sciences*.

4. Rao, S., et al. (2017). A forward model for electrical conduction in soil-root continuum: a virtual rhizotron study. *4th International Workshop on Geoelectrical Monitoring*.

5. Rao, S., et al. (2017). Characterizing root system characteristics with Electrical resistivity Tomography: a virtual rhizotron simulation. *EGU General Assembly Conference Abstracts*.

Peer Review & Professional Service

- Reviewer for *Nordic Machine Intelligence*
- Reviewer for *Plants in Silico*
- Reviewer for *PeerJ Computer Science*
- Reviewer for *Plant and Soil*
- Reviewer for *Vadose Zone Journal*
- Reviewer for *Agricultural Research*

Technical Skills

Geospatial & Remote Sensing: Google Earth Engine, Sentinel-1 C-band SAR processing, GDAL/OGR, Shapefile/GeoJSON manipulation, Land Resource Inventory (LRI) spatial

filtering, Stratified spatial sampling design, SRTM terrain correction, Electrical Resistivity Tomography (ERT), Time Domain Reflectometry (TDR), Ground Penetrating Radar (GPR)

Statistical & Causal Methods: Difference-in-Differences (DiD), Before–After Control–Impact (BACI) design, Event-study analysis, TOST equivalence testing, Bootstrap resampling, OLS with robust standard errors, Exponential decay modeling

Programming and Scripting: Python (NumPy, Pandas, Plotly, Matplotlib), C++, FORTRAN, MATLAB, JavaScript (Vue.js, Chart.js), HTML/CSS, PyQt (GUI development)

Machine Learning and Data Science: Scikit-learn, TensorFlow, PyTorch, Keras, Random Forests, LSTM neural networks, Support Vector Machines, Computer Vision (OpenCV), Cross-correlation, Anomaly Detection, Seasonal Decomposition, Time-series Forecasting

Software Engineering: Docker, Git (version control), CI/CD workflows, WebSockets, API integration (financial data, remote sensing data), Hugo (static site development), Data Pipelines

High-Performance Computing: MPI, OpenMP, HPC clusters, Linux systems, Parallel Computing, Simulation Scaling

Modeling and Simulation: Digital Twin Development, Model Coupling (crop and root models, loose coupling methods), Numerical Methods, Finite Element Modeling (FEM), PIC Plasma Simulations, Hydrological and Biophysical Simulation Pipelines

Data Visualization: Interactive Dashboards (Plotly, Vue.js, Leaflet, Chart.js), Scientific Animations, Visualization for Remote Sensing Data, Custom Visualization for Soil–Moisture–Temperature Dynamics

Instrumentation and Experimental: Satellite Remote Sensing (SAR, LAI, backscatter analysis), Geoelectric Methods, Atomic Force Microscopy (AFM), Optical Test Benches, Lock-in Amplifiers, Field Data Collection Protocols

Science Communication and Media: Technical Writing, Video Editing (Final Cut Pro, DaVinci Resolve), 2D Animation (Pencil2D), Educational Content Creation (YouTube – Compute Stories)

Professional Development

- Land in Focus - Basics of Remote Sensing, EO College November 2022
- Advanced HPC training workshops and computational modeling courses
- Intro to Reinforcement Learning, University of Alberta, CoursEra, Aug 2025

Volunteer Experience

Blog Committee Member — Young Hydrologic Society (YHS) April 2025 – Present

- Selected to contribute to the YHS Blog Committee, creating educational content for early career hydrologists.
- Develop blog posts on topics related to computational hydrology, digital twins, and environmental modeling.
- Support the YHS mission of promoting participation of early career researchers in the hydrologic science community.